Pooling Correlation Coefficients

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Introduction

Pearson correlation coefficients are bounded to $[-1, 1]$, and their sampling distributions are skewed with heteroscedastic variance — the variance depends on the underlying ρ and on n. These properties make raw correlations unsuitable for inverse-variance meta-analysis. Fisher's z transformation converts each correlation to an approximately Normal scale with variance independent of ρ, enabling standard random-effects pooling. After pooling on the z scale, the result is back-transformed to a correlation for interpretation.

Prerequisites

A working understanding of Pearson correlation, variance-stabilising transformations, and inverse-variance meta-analysis under random effects.

Theory

The Fisher z transformation is

$$z = \tfrac{1}{2}\log\left(\frac{1+r}{1-r}\right) = \tanh^{-1}(r), \qquad \mathrm{Var}(z) \approx \frac{1}{n-3}.$$

The variance is constant (depends only on n, not ρ), so the inverse-variance weighting in meta-analysis is well-defined. Pool z values across studies via fixed-effect or random-effects meta-analysis, then back-transform $r = \tanh(z)$ for reporting on the natural correlation scale.

The back-transformation preserves coverage at the original scale provided the transformed pooling is correctly weighted; reporting both z (with SE) and r (with CI) is standard.

Assumptions

Bivariate-Normal data in each study (the basis of the variance formula); independent studies; sufficient sample size per study ($n > 10$ as a rough minimum) for the asymptotic distribution to be reasonable.

R Implementation

```r
library(metafor)

set.seed(2026)
k <- 10
# Simulated correlations and sample sizes
df <- data.frame(
  ri = c(0.25, 0.32, 0.18, 0.40, 0.28, 0.35, 0.22, 0.30, 0.45, 0.38),
  ni = sample(50:300, k, replace = TRUE)
)

es <- escalc(measure = "ZCOR", ri = ri, ni = ni, data = df)
```

```r
re <- rma(yi, vi, data = es, method = "REML")

# Convert pooled z back to correlation
pooled_z  <- re$b
pooled_r  <- tanh(pooled_z)
ci_r      <- tanh(c(re$ci.lb, re$ci.ub))

c(pooled_r = round(pooled_r, 3),
  lower = round(ci_r[1], 3), upper = round(ci_r[2], 3))
re$I2
```

Output & Results

`escalc(measure = "ZCOR")` computes Fisher z and its variance per study; `rma()` fits the random-effects model on the z scale. The pooled correlation is approximately 0.31 with a tight 95 % CI; I^2 quantifies heterogeneity.

Interpretation

A reporting sentence: "Random-effects meta-analysis of 10 studies (REML, n total = 1{,}523) gave a pooled correlation of $r = 0.31$ (95 % CI 0.25 to 0.37) with modest heterogeneity ($I^2 = 28\%$, $\tau^2 = 0.012$); pooling on the Fisher z scale and back-transforming maintains correct coverage despite the $[-1, 1]$ bound on r." Always pool on the z scale and report both Fisher z and back-transformed r.

Practical Tips

- Always pool on the Fisher z scale; pooling raw r biases estimates for high correlations and produces incorrect coverage.
- `metafor::escalc(measure = "ZCOR")` handles the Fisher z transformation and variance computation in a single call.
- Report both z (with SE) and back-transformed r (with CI); r is more interpretable for substantive readers.
- For partial correlations, apply the same approach with the degrees-of-freedom-adjusted sample size $n - p$ where p is the number of partialled-out variables.
- Hartung-Knapp small-sample adjustment (`rma(..., test = "knha")`) is especially important when studies are few or unevenly sized; standard inverse-variance gives over-confident intervals in those settings.
- For meta-analysis of intraclass correlations or other correlation-like statistics, the same Fisher z framework applies with appropriate variance formulae.

R Packages Used

`metafor` for `escalc()`, `rma()`, and the canonical correlation-pooling workflow; `meta` for an alternative interface with cleaner forest-plot output; `psychmeta` for psychometric meta-analysis with Hunter-Schmidt artefact corrections; `clubSandwich` for robust variance estimation in correlated-effect meta-analyses.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Systematic reviews and PRISMA
- Meta-analysis basics
- Network meta-analysis